

1. Funds and Backtest Design

1.1 NexaVest Fund Range

NexaVest is a systematic multi-asset investment platform for financially literate investors who wish to compare investment strategies independently. It focuses on transparent portfolio rules, risk characteristics and out-of-sample evidence. The platform covers Equity, Crypto and Combined equity–crypto funds.

Four portfolio methods—Equal Weight, Minimum Variance, Maximum Sharpe and Risk Parity—are applied across the three asset families, producing 12 base funds. Two additional sentiment-enhanced Minimum Variance funds are later used to test whether industry news sentiment can add value to traditional portfolios.

Asset Family	Equal Weight	Min Variance	Max Sharpe	Risk Parity
Equity	✓	✓	✓	✓
Crypto	✓	✓	✓	✓
Combined	✓	✓	✓	✓

Table 1. NexaVest Fund Architecture

Note: Combined funds contain both equity and cryptocurrency assets. Two sentiment-enhanced Minimum Variance funds are evaluated separately in the innovation section.

1.2 Portfolio Construction and Investment Rules

The four methods represent different investment objectives. Equal Weight provides a transparent benchmark. Minimum Variance seeks lower portfolio volatility, while Maximum Sharpe targets higher expected return per unit of risk under a zero risk-free-rate assumption. Risk Parity focuses on distributing portfolio risk more evenly across assets.

All funds are long-only and fully invested. Combined funds also impose a 25% cap on total cryptocurrency exposure to prevent high crypto volatility from dominating the portfolio while retaining potential diversification benefits.

A turnover-based transaction cost of 10 basis points is applied at each rebalance. Reported fund performance therefore reflects net returns after estimated trading costs rather than frictionless performance.

1.3 Walk-Forward Out-of-Sample Design

NexaVest uses a walk-forward out-of-sample (OOS) backtest to reduce look-ahead bias. At each rebalance date, portfolio weights are estimated using only information available before that date and are then applied to the following holding period.

Equity and Combined funds use a 252-observation estimation window and rebalance every 21 trading observations, while Crypto funds use a 365-observation window and rebalance every 30 calendar observations. The different settings reflect the different trading calendars of equities and cryptocurrencies, with annualisation factors of 252 and 365 respectively. A total of 436 rebalance records were audited. All training periods ended before their corresponding rebalance dates, and

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all optimisation runs completed successfully without fallback. These checks provide additional evidence that the backtest is look-ahead safe and technically stable.

Fund Family	Estimation Window	Rebalance Step	Annualisation	First OOS Date	Last OOS Date	OOS Observations
Equity	252 observations	21 trading observations	252	4 Jan 2021	29 Dec 2023	753
Combined	252 observations	21 trading observations	252	4 Jan 2021	29 Dec 2023	753
Crypto	365 observations	30 calendar observations	365	1 Jan 2021	31 Dec 2023	1,095

Table 2. Walk-Forward Out-of-Sample Backtest Design

2. Out-of-Sample Fund Results and Fact Sheets

2.1 Performance Comparison

Fund	Annualised Return	Volatility	Sharpe	Max Drawdown
Equity Equal Weight	12.61%	16.17%	0.815	-20.32%
Equity Minimum Variance	6.23%	12.57%	0.543	-15.39%
Equity Maximum Sharpe	8.83%	16.61%	0.593	-18.81%
Equity Risk Parity	9.85%	14.59%	0.717	-18.43%
Crypto Equal Weight	34.14%	80.42%	0.772	-81.60%
Crypto Minimum Variance	62.05%	72.40%	1.031	-73.20%
Crypto Maximum Sharpe	21.27%	77.65%	0.638	-81.85%
Crypto Risk Parity	37.77%	78.30%	0.805	-79.97%
Combined Equal Weight	14.94%	21.25%	0.762	-28.75%
Combined Minimum Variance	6.34%	12.59%	0.551	-15.53%
Combined Maximum Sharpe	15.99%	19.82%	0.847	-19.02%
Combined Risk Parity	13.94%	16.01%	0.895	-19.72%

Table 3. Out-of-Sample Performance across NexaVest Funds

Out-of-sample results show that there are marked differences in risk-return profiles among the three fund categories. Crypto funds achieved higher overall annualised returns, but at the same time incurred significantly higher volatility and drawdowns than equity and Combined funds. In particular, the Crypto Minimum Variance fund achieved an annualised return of 62.05 per cent and a Sharpe ratio of 1.031, representing the highest risk-adjusted return among the benchmark funds; however, its annualised volatility of 72.40 per cent and maximum drawdown of -73.20 per cent also indicate that such high returns are accompanied by extremely high investment risk.

By comparison, the Combined funds offer a more balanced risk-return profile. The Combined Maximum Sharpe fund achieved an annualised return of 15.99 per cent, whilst the Combined Risk Parity fund recorded an annualised return of 13.94 per cent, a volatility of 16.01 per cent and

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a Sharpe ratio of 0.895. Although the latter did not achieve the highest absolute return, it struck a good balance between return, risk and risk-adjusted performance.

The Equity funds as a whole carry the lowest risk. Among them, the Equity Minimum Variance fund has a volatility of just 12.57 per cent and a maximum drawdown of -15.39 per cent, but its annualised return is also relatively low at 6.23 per cent. This illustrates that whilst the objective of reducing volatility can help control risk, it may also come at the expense of some potential upside returns.

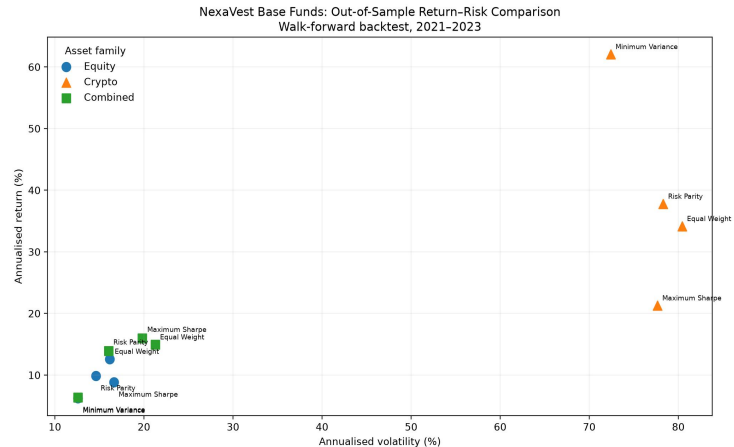


Figure 1. Out-of-Sample Return-Risk Comparison across NexaVest Base Funds

Figure 1: Crypto is all in the upper right corner, with high returns and high volatility. Equity is in the lower left area. Combined mostly falls somewhere in between. This indicates that a multi-asset portfolio may capture some of the return potential of crypto without bearing the same level of risk as a crypto-only portfolio.

2.2 Growth of \$1 and Drawdown

Growth of \$1 further illustrates the cumulative performance of the different portfolio strategies over the entire out-of-sample period. The Combined Maximum Sharpe strategy ultimately achieved the highest cumulative wealth growth, corresponding to an annualised return of 15.99 per cent. The Combined Risk Parity strategy ended up with a slightly lower final wealth level, but its lower volatility resulted in a higher Sharpe ratio. By contrast, the Combined Minimum Variance strategy recorded the slowest wealth growth, which is consistent with its objective of reducing portfolio volatility.

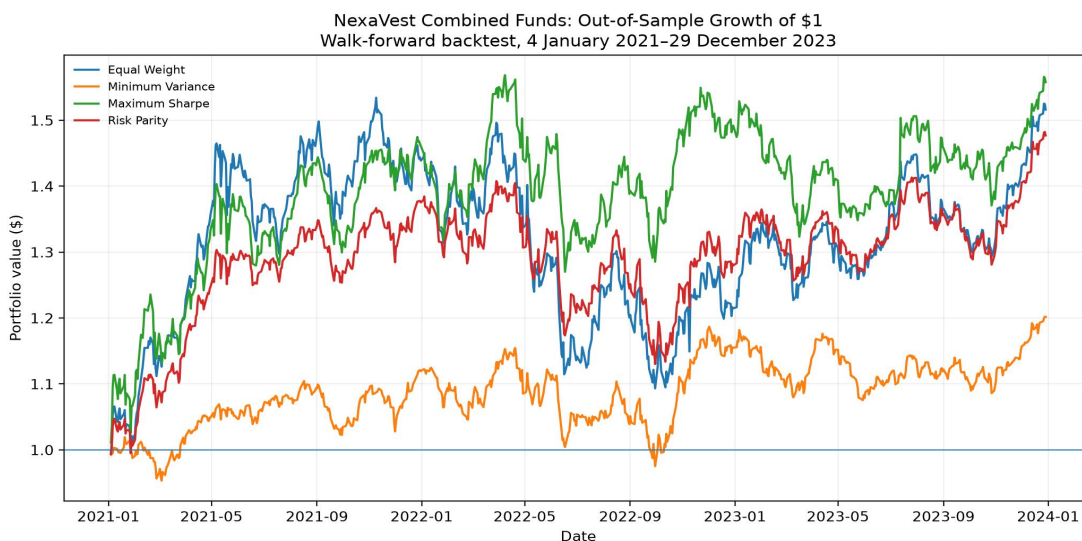


Figure 2. Out-of-Sample Growth of \$1 for NexaVest Combined Funds

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The Drawdown chart offers a different risk perspective from the return chart. The maximum drawdown of Combined Minimum Variance is approximately -15.53%, which is the shallowest among the four Combined strategies, while the maximum drawdown of Equal Weight reaches -28.75%. This indicates that higher cumulative returns do not necessarily mean better downside protection. Therefore, when comparing funds, investors need to consider both returns and the risk of capital loss simultaneously.

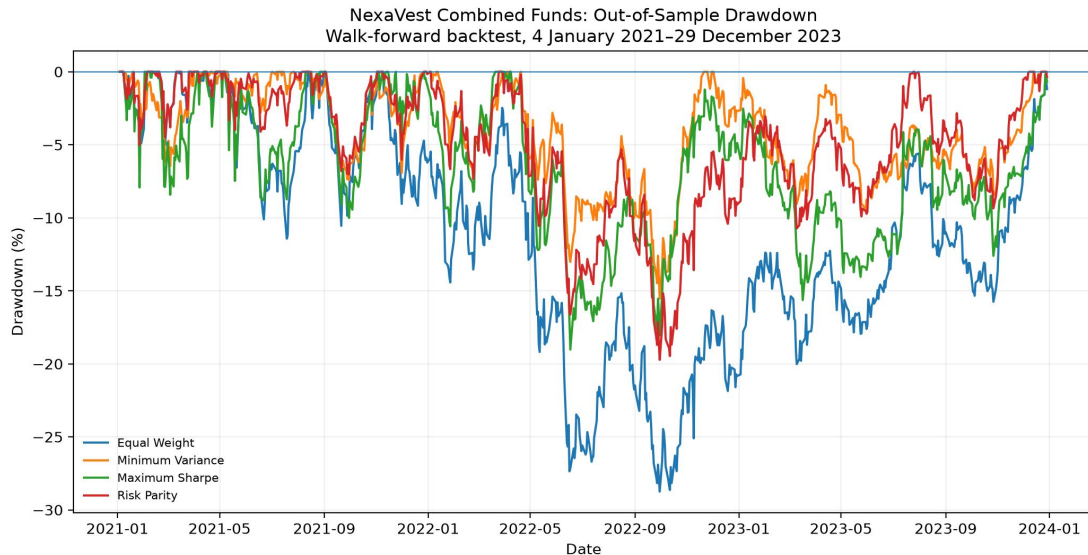


Figure 3. Out-of-Sample Drawdowns for NexaVest Combined Funds

2.3 Portfolio Weights and Fund Fact Sheets

Portfolio weightings further illustrate why different strategies yield different risk-return outcomes. The Combined fund's cryptocurrency allocation is always subject to a 25 per cent cap, but different optimisation methods utilise this risk budget to varying degrees. Meanwhile, the Risk Parity fund's equity allocation adjusts in line with changes in estimated risk, rather than maintaining a fixed weighting. This demonstrates that NexaVest's systematic funds generate different risk exposures through rule-driven, dynamic allocation.

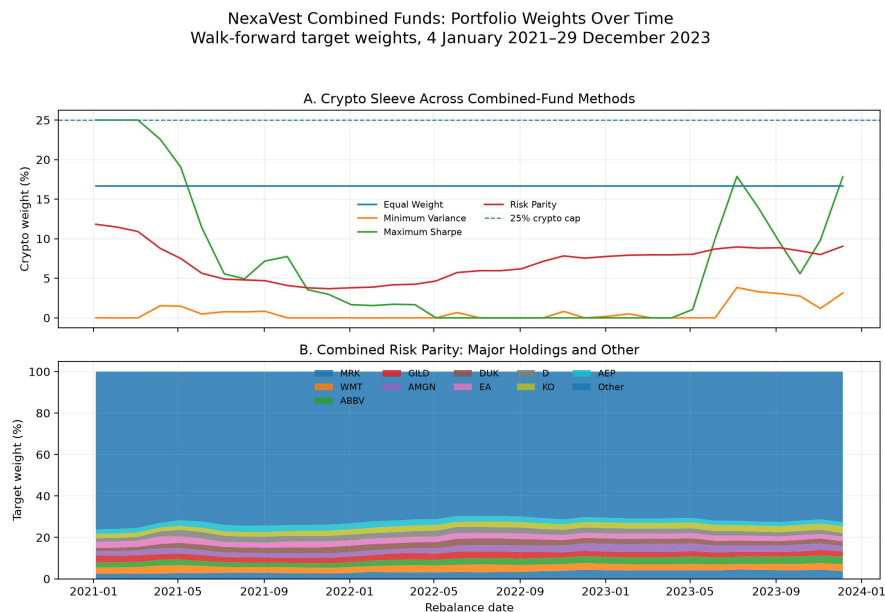


Figure 4. Portfolio Weights

3. Sector Sentiment Index

3.1 Finance-Aware VADER

NexaVest uses VADER to assign sentiment scores to stock news headlines (Hutto & Gilbert, 2014). As the data comprises only news headlines rather than full articles, the sentiment scores should be regarded as an approximate indicator of market sentiment rather than a comprehensive measure of investors' actual sentiment.

As the standard VADER has limited ability to recognise certain financial terms, this project has expanded the original dictionary by adding 25 financial terms. For example, expressions related to 'earnings', 'upgrade', 'downgrade' and 'financial weakness' have been assigned sentiment scores that are more appropriate to a financial context. The purpose of this expansion is not to directly improve investment returns, but to reduce the number of instances where financial news is incorrectly classified as neutral, and to enhance the sentiment indicator's coverage of financial language.

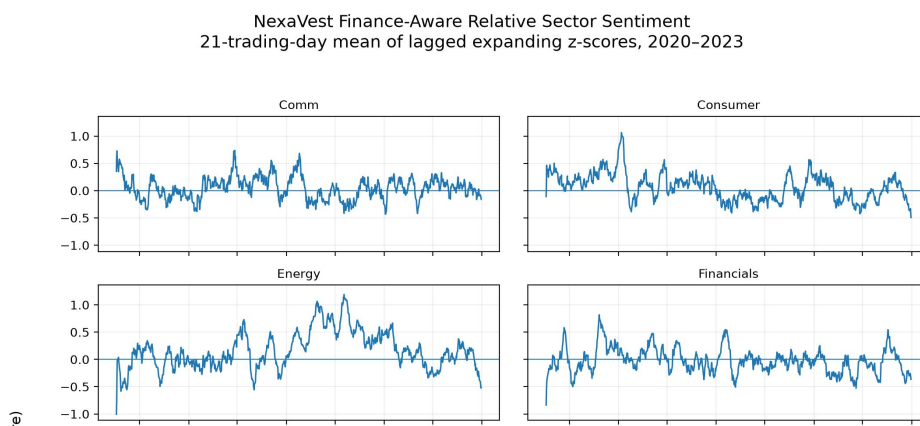
3.2 Sector Sentiment Construction

News items are first scored at the headline level, then aggregated to form a ticker-day sentiment. Ticker-day sentiment is then equal-weighted across stocks within each sector to construct daily sentiment indices for the 10 equity sectors.

For ticker-days with no news, this project sets the sentiment value to 0, whilst retaining the coverage ratio to record the proportion of shares within a sector that actually had news coverage on that day. This approach ensures a complete industry time series whilst distinguishing between situations where sentiment is 'close to neutral' and those where 'news coverage is insufficient'.

To prevent future information from influencing investment decisions, industry sentiment signals are delayed by at least one trading day before being applied to investment portfolios. NexaVest further standardises industry sentiment using historical data, ensuring that the signal reflects the current sentiment within a given industry relative to its own historical levels, rather than directly comparing raw VADER scores across different sectors.

As can be seen from the industry time series, relative sentiment across different sectors changes markedly over time, and sentiment shocks do not typically persist for long. This suggests that news sentiment is better suited as dynamic supplementary information rather than a long-term, fixed metric for asset ranking.



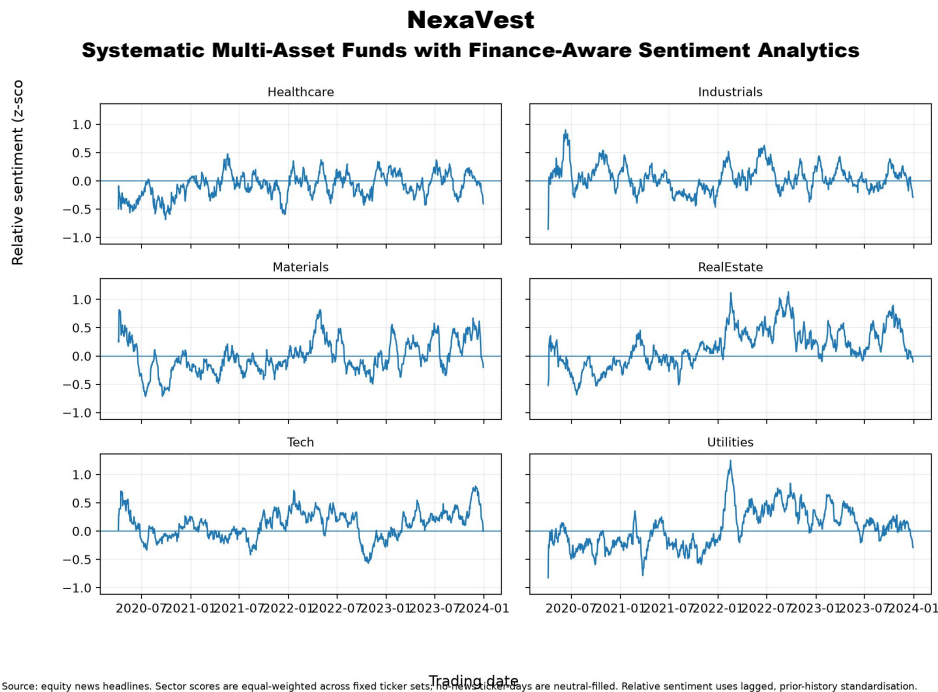


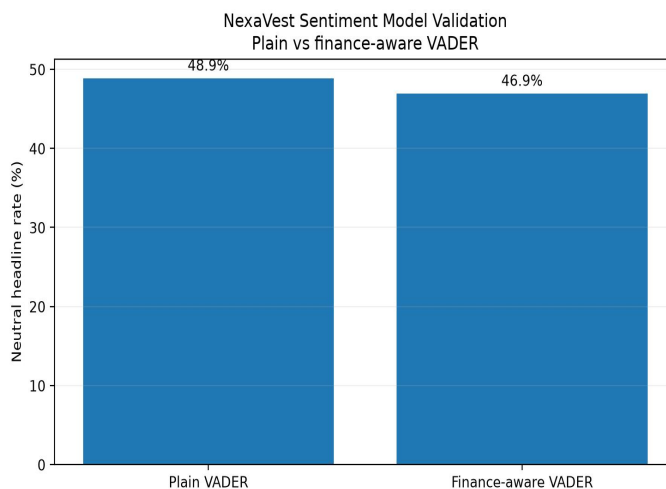
Figure 5. NexaVest Finance-Aware Relative Sector Sentiment

3.3 Sentiment Model Validation

The model validation results show that the standard VADER classifies approximately 48.9 per cent of financial news headlines as neutral, whereas, following the incorporation of the financial lexicon, the proportion of neutral classifications fell to approximately 46.9 per cent. A total of 8,741 headline scores were altered as a result of the financial lexicon expansion. This suggests that the finance-aware lexicon captures a broader range of financially relevant language.

However, the decrease in the proportion of neutral ratings does not prove that sentiment signals possess predictive power. It merely indicates that the model's coverage of financial terminology has improved. Consequently, the sector sentiment index is initially regarded as an independent market analysis tool within NexaVest, and its investment value requires further evaluation through subsequent out-of-sample testing of sentiment – portfolio fusion.

Finance-aware VADER has reduced the neutral classification rate, indicating that expanding the dictionary has improved coverage of financial language. However, this result merely confirms an improvement at the text classification level and does not imply that the model is yet capable of investment forecasting.



Source: 105,329 unique equity-news headlines. A lower neutral rate indicates broader finance vocabulary coverage, not evidence of return predictability.

Figure 6. Plain versus Finance-Aware VADER Validation

4. Extensions and Innovations

4.1 Expanded Fund Range and Risk Controls

The first expansion of NexaVest is to expand the basic requirements from a single Combined fund to three types of assets: Equity, Crypto, and Combined, and apply four portfolio methods respectively. In addition to increasing investment options, this design also allows for the comparison of the performance of the same optimization method across different asset classes.

In addition, the Combined fund has set a 25% cap on the weight of cryptocurrencies and adds 10 basis points of transaction costs at each rebalancing. The former is used to limit the impact of highly volatile crypto assets on the overall portfolio, while the latter makes out-of-sample results closer to the actual investment environment.

4.2 Coverage-Aware Sentiment Signal

The second extension involves incorporating a financial lexicon into the traditional VADER model and taking news coverage into account. For ticker-days with no news, the sentiment value is set to 0, but the system retains the coverage ratio to distinguish between ‘neutral sentiment’ and ‘lack of news’.

In portfolio applications, NexaVest utilises a coverage-adjusted sentiment signal. When news coverage for a particular sector is low, the influence of its sentiment signal on portfolio weightings is correspondingly reduced. This design mitigates the risk of a small amount of news exerting an excessive influence on the overall sector allocation.

4.3 Sentiment–Portfolio Fusion

To test whether unstructured news information can enhance traditional investment portfolios, this project incorporated a sentiment tilt into both the Equity Minimum Variance and Combined Minimum Variance funds.

On each rebalancing date, the system adjusts equity weightings using sector-specific relative sentiment signals that are at least one trading day lagged. Positive sector sentiment increases the relative weighting of the corresponding stocks, whilst negative sentiment reduces their weighting. The intensity of the sentiment adjustment was preset at 0.30 and remained fixed until the final out-of-sample results were available, in order to minimise the risk of data mining arising from retroactively adjusting parameters based on the results.

For the Combined fund, the sentiment tilt was applied only to the equity component, whilst the cryptocurrency weightings remained unchanged. Consequently, the sentiment model did not alter the original cryptocurrency allocation budget, nor did it exceed the 25 per cent cryptocurrency cap.

4.4 Before-and-After Evaluation

Out-of-sample results indicate that incorporating sentiment signals did not improve the risk-adjusted performance of the Minimum Variance fund. The Sharpe ratio for the Equity Minimum Variance fund fell from approximately 0.543 to 0.518, whilst that of the Combined

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Minimum Variance fund fell from approximately 0.551 to 0.526. The annualised returns of both sentiment-enhanced funds declined by approximately 0.3 percentage points, whilst their maximum drawdowns widened slightly.

Fund	Annualised Return	Volatility	Sharpe	Max Drawdown
Equity Min Variance	6.23%	12.57%	0.543	-15.39%
Equity Min Variance + Sentiment	5.91%	12.63%	0.518	-16.58%
Combined Min Variance	6.34%	12.59%	0.551	-15.53%
Combined Min Variance + Sentiment	6.02%	12.65%	0.526	-16.73%

Table 4. Sentiment Fusion Before-and-After Comparison

Although this result is negative, it is nonetheless of analytical value. It illustrates that richer information does not necessarily translate automatically into better investment performance. News headlines contain noise, sector sentiment coverage is inconsistent, and sentiment tilts can increase portfolio turnover and therefore transaction costs.. Therefore, sentiment is better suited as a supplementary signal requiring further validation, rather than a direct substitute for traditional risk allocation methods.

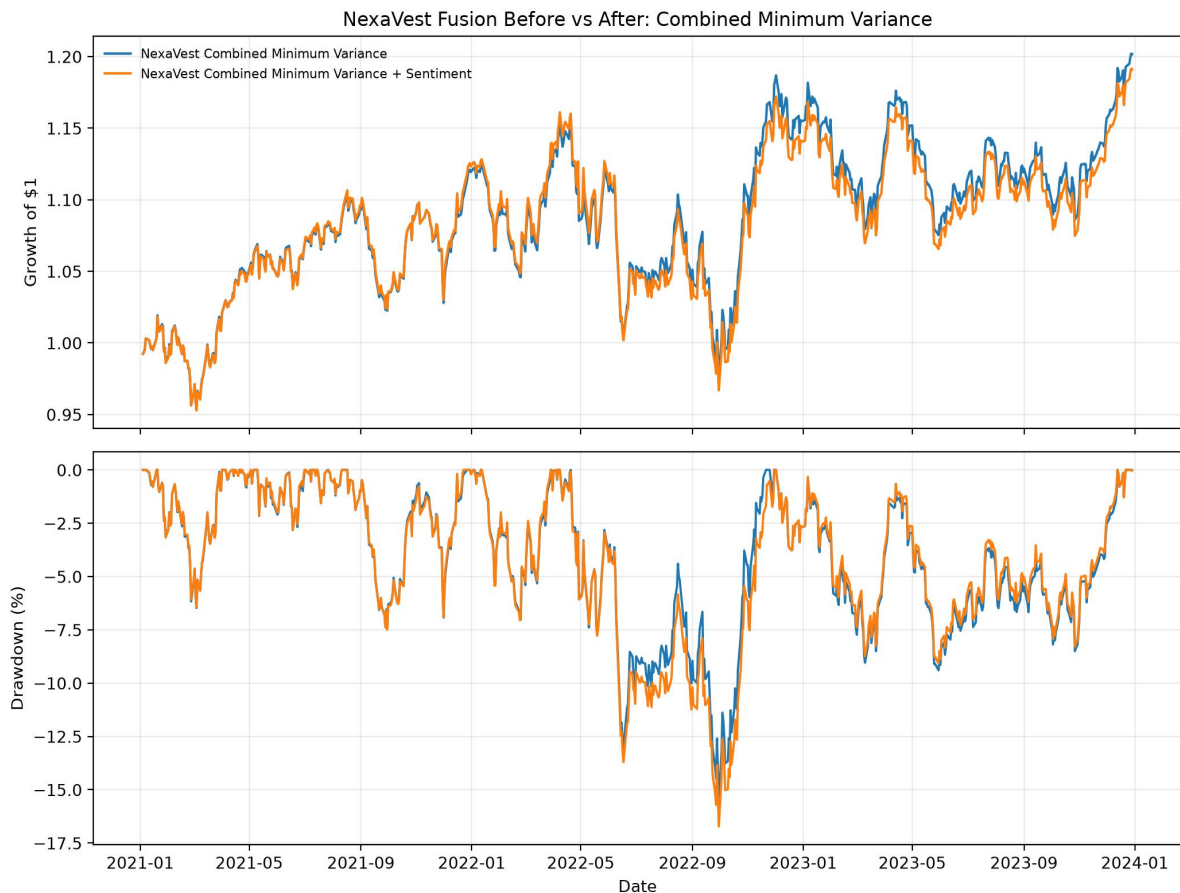


Figure 7. Sentiment Fusion: Before-and-After Growth and Drawdown

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5. NexaVest App and Investor Journey

NexaVest is presented through an interactive Streamlit dashboard designed for financially literate investors who want to compare systematic investment strategies and understand their risk–return characteristics.

The investor journey follows four main steps. First, users compare Equity, Crypto and Combined funds across different portfolio methods. Second, they open an individual fund fact sheet to review out-of-sample performance, Growth of \$1, risk measures and latest target holdings. Third, users can construct their own allocation across selected NexaVest funds. Finally, the Sentiment Lens allows investors to explore sector-level news sentiment and its historical evolution.

The application uses only pre-computed portfolio and sentiment results rather than rerunning optimisation or sentiment analysis during deployment. This improves speed, reproducibility and consistency between the report and the investor-facing dashboard.

Overall, the app converts the quantitative analysis into a more accessible decision-support tool. Rather than recommending a single “best” fund, NexaVest allows investors to compare return potential, volatility, drawdown and portfolio composition according to their own risk preferences.

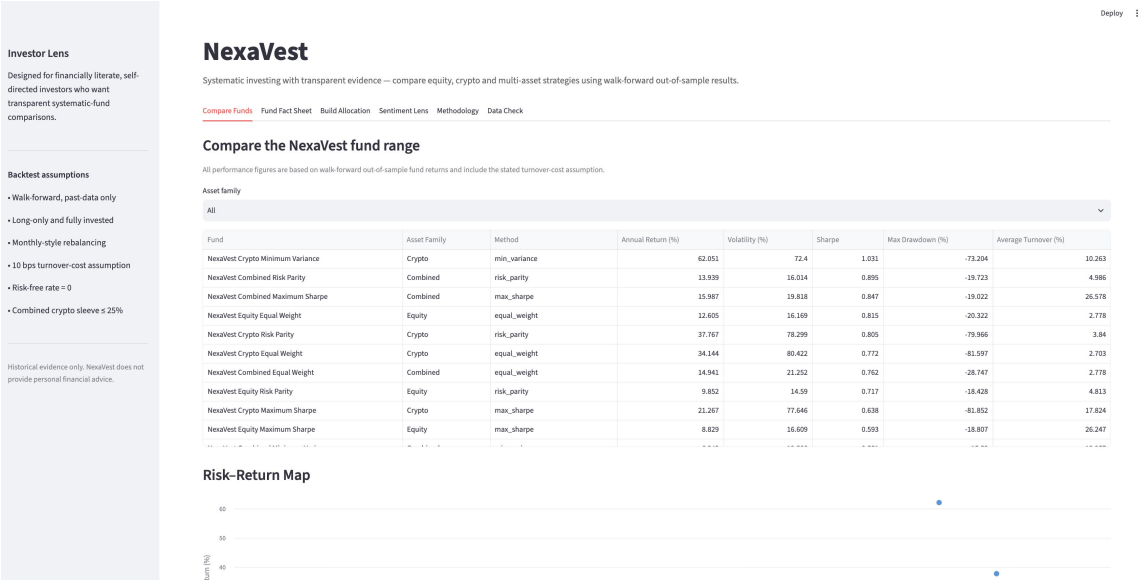


Figure 8. NexaVest Investor Dashboard and Customer Journey

6. Critical Reflection and Recommendations

The project demonstrates that systematic portfolio construction can generate clearly different risk–return profiles across asset classes and investment methods. However, the results also show that model complexity does not automatically produce better out-of-sample performance. For example, Maximum Sharpe optimisation was not always the strongest strategy in realised Sharpe terms, while the sentiment-enhanced Minimum Variance funds slightly underperformed their base portfolios.

Three improvements are therefore recommended.

First, future versions should test portfolio parameters across multiple rolling or expanding out-of-sample periods rather than relying on a single historical evaluation window. This would provide stronger evidence on whether strategy performance is stable across different market regimes.

Second, the sentiment model should be validated using a manually labelled financial-news sample. The current finance-aware VADER extension improves financial-language coverage, but lower neutral classification does not prove greater sentiment accuracy or return predictability. A labelled validation sample would allow more direct assessment of classification quality.

Third, future sentiment fusion should consider signal strength, news coverage and transaction costs jointly. The current results suggest that sentiment tilts can increase turnover without improving risk-adjusted returns. A more selective approach, where portfolio weights are adjusted only when sentiment signals are sufficiently strong and well covered, may reduce unnecessary trading.

Overall, NexaVest should therefore be viewed as a transparent decision-support platform rather than a system that identifies one universally optimal strategy. Its main value lies in allowing investors to compare systematic investment choices using consistent out-of-sample evidence.

References

Hutto, C. J., & Gilbert, E. (2014). VADER: A parsimonious rule-based model for sentiment analysis of social media text. *Proceedings of the International AAAI Conference on Web and Social Media*, 8(1), 216–225.
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